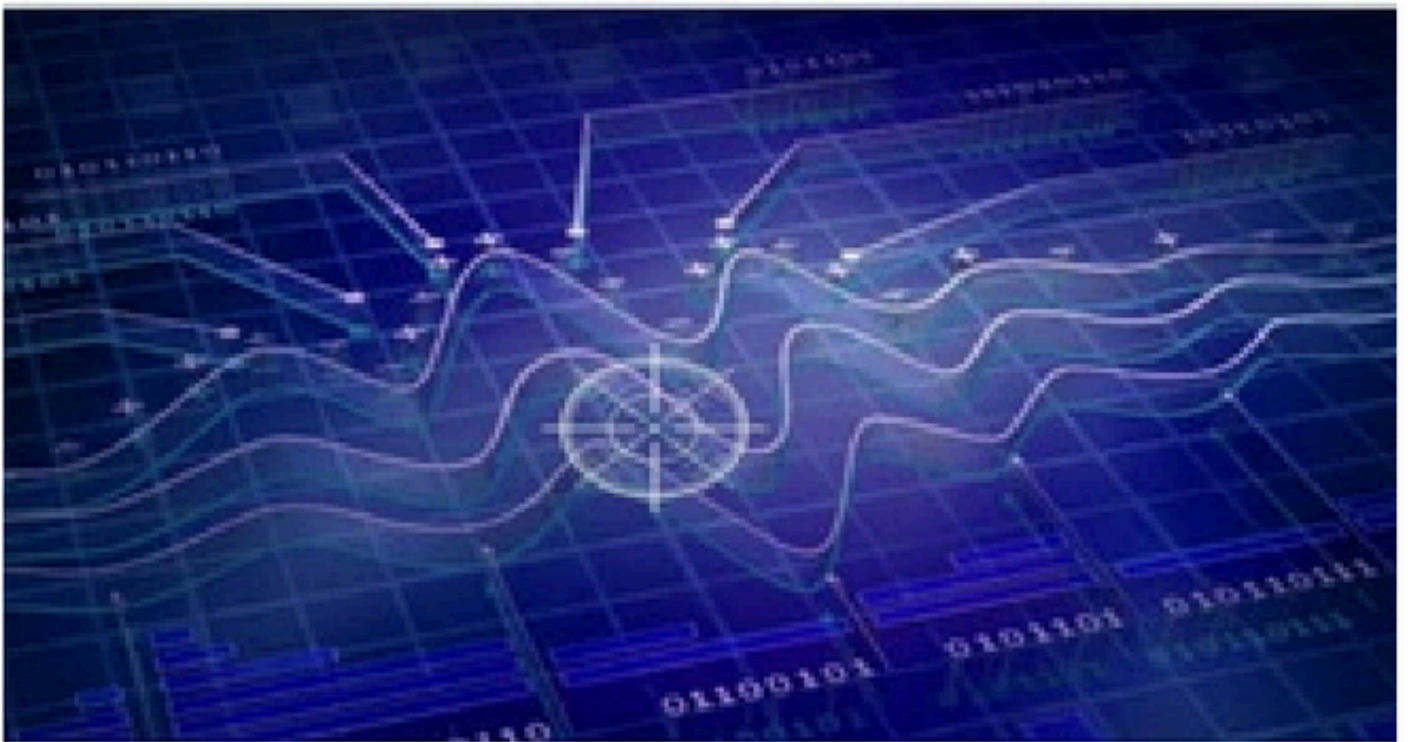


DIGITAL SIGNAL PROCESSING LAB

Experiment-6: DTMF (Dual Tone Multifrequency or TouchTone) coder / decoder



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Objectives:

Study and Analysis of DTMF (Dual Tone Multifrequency, or TouchTone) coder / decoder using Digital FIR Filter in MATLAB.

Will learn:

- (a) Functionality of DTMF and its different applications.
- (b) Implementation of BandPass Digital Filters.
- (c) DTMF Coder/Decoder.
- (d) How to characterize a filter by knowing how it reacts to different frequency components in the input.

```
clc;  
clear;  
close all;
```

Defining dtmf parameters

```
Fs = 8000; % Sampling frequency  
dtmf_rows = [697,770,852,941];  
dtmf_cols = [1209,1336,1477,1633];  
dtmf_freq = [dtmf_rows dtmf_cols];  
dtmf_keys = ['1', '2', '3', 'A';  
             '4', '5', '6', 'B';  
             '7', '8', '9', 'C';  
             '*', '0', '#', 'D'];
```

Signal generation for key '5'(example)

```
t = 0:1/Fs:0.5-1/Fs;% duration  
key_row = 2; % Row frequency index for '5'  
key_col = 2; % Column frequency index for '5'  
x = sin(2*pi*dtmf_rows(key_row)*t) + sin(2*pi*dtmf_cols(key_col)*t); % dtmf tone  
sound(x, Fs); % Play the tone
```

FIR bandpass filter parameters

```
L = 101; % odd for symmetry  
BW = 50; % bandwidth(Hz)  
n = 0:(L-1); %discrete time vector
```

```
% Hann window
```

```
win = 0.5 - 0.5*cos(2*pi*n/(L-1));
```

Filter bank design and signal filtering

```
figure;
```

```
for k = 1:length(dtmf_freq)
```

```
    fc = dtmf_freq(k); % desired center frequency
```

```
    wc = 2*pi*fc/Fs;
```

```
    % Lowpass impulse response (sinc function)
```

```
    h_lowpass = (2*BW/Fs) * sinc(2*BW/Fs * (n-(L-1)/2));
```

```
    % Modulate to center frequency
```

```
    h_bandpass = h_lowpass .* cos(wc * (n-(L-1)/2));
```

```
    % Apply window
```

```
    h_bandpass = h_bandpass .* win;
```

```
    % Normalize frequency response maximum to one
```

```
    h_bandpass = h_bandpass / sum(abs(h_bandpass));
```

```
    % DFT for filter frequency response
```

```
    H = zeros(1, 1024);
```

```
    for m = 1:1024
```

```
        w = 2*pi*(m-1)/1024;
```

```
        for j = 1:L
```

```
            H(m) = H(m) + h_bandpass(j) * exp(-1i*w*(j-1));
```

```
        end
```

```
    end
```

```
    plot(linspace(0,Fs,1024), abs(H), 'LineWidth', 1.5); hold on;
```

```
end
```

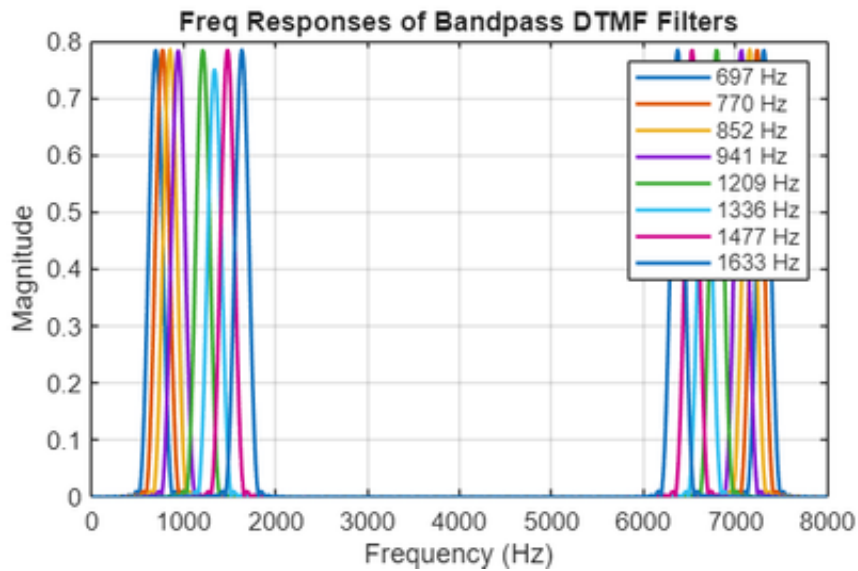
```
xlabel('Frequency (Hz)');
```

```
ylabel('Magnitude');
```

```
title('Freq Responses of Bandpass DTMF Filters');
```

```
legend(arrayfun(@(f) sprintf('%d Hz',f),dtmf_freq,'UniformOutput',false));
```

```
grid on;
```



Filter the outputs

```

filtered_outputs = zeros(length(dtmf_freq), length(x));

for k = 1:length(dtmf_freq)
    fc = dtmf_freq(k); % desired center frequency
    wc = 2*pi*fc/Fs;

    % Lowpass impulse response (sinc function)
    h_lowpass = (2*BW/Fs) * sinc(2*BW/Fs * (n-(L-1)/2));

    % Modulate to center frequency
    h_bandpass = h_lowpass .* cos(wc * (n-(L-1)/2));

    % Apply window
    h_bandpass = h_bandpass .* win;

    % Normalize frequency response maximum to one
    h_bandpass = h_bandpass / sum(abs(h_bandpass));

    % convolution
    filtered = zeros(size(x));

    for i = 1:length(x)
        for j = 1:L
            if i-j+1 > 0
                filtered(i) = filtered(i) + h_bandpass(j) * x(i-j+1);
            end
        end
    end
end

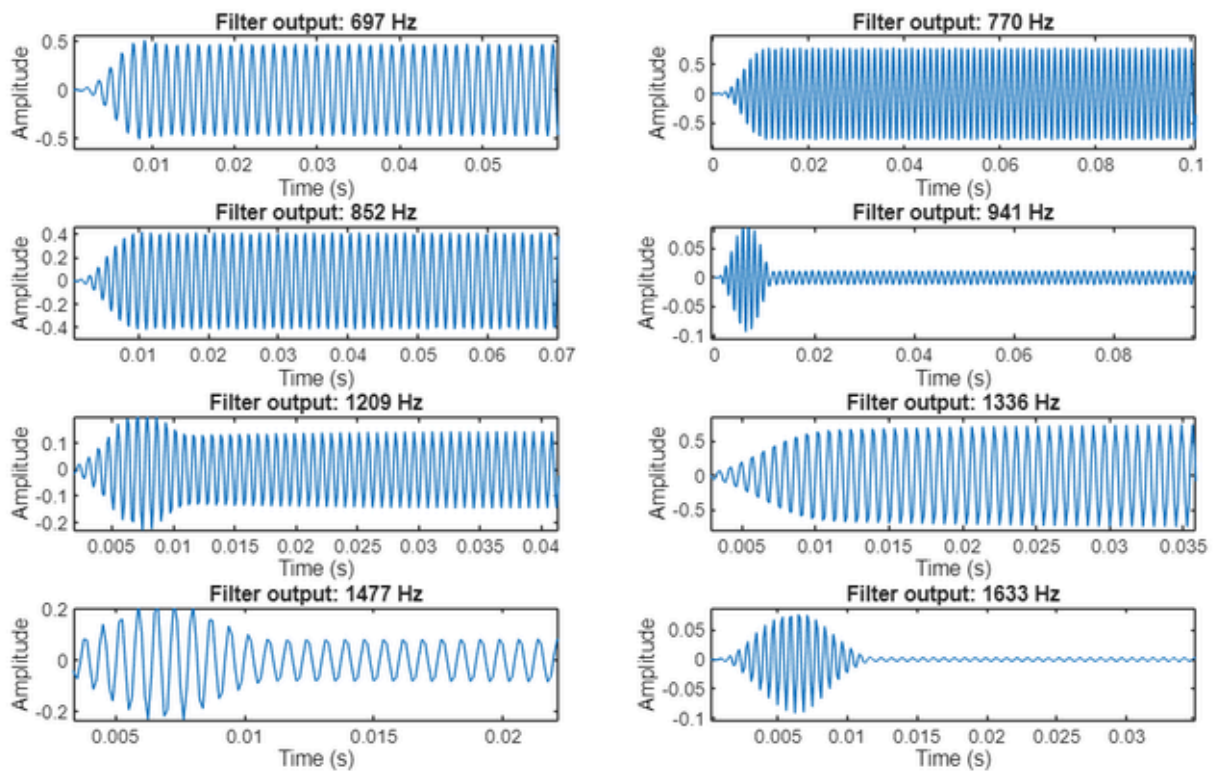
```

```

        filtered_outputs(k, :) = filtered;
    end

figure;
for k = 1:length(dtmf_freq)
    subplot(4,2,k);
    plot(t, filtered_outputs(k,:));
    title(sprintf('Filter output: %d Hz', dtmf_freq(k)));
    xlabel('Time (s)');
    ylabel('Amplitude');
end

```



DTMF key detection

```

signal_len = length(x);
energy = max(abs(filtered_outputs(:,1:signal_len)), [], 2); % Peak measure per filter
[~, idxs] = sort(energy, 'descend');
row_idx = idxs(find(idxs<=4, 1)); % first index in 1..4

```

```
col_idx = idxs(find(idxs>4, 1)) - 4; % first index in 5..8 (adjust for col)

detected_key = dtmf_keys(row_idx, col_idx);

fprintf('Detected key: %s\n', detected_key);

Detected key: 5
```

Results and Observations

1. Frequency Response Analysis: The frequency response plots revealed sharp, well-defined peaks at each of the eight DTMF frequencies. An important observation was the appearance of symmetric peaks at both low frequencies (697-1633 Hz) and high frequencies (6367-7303 Hz). This symmetry is a fundamental property of real-valued FIR filters, where the frequency response exhibits conjugate symmetry around the Nyquist frequency (4000 Hz for $F_s = 8000$ Hz). While these mirror images are mathematically expected, they do not interfere with DTMF detection since all signal components lie in the lower frequency range.

2. Signal Filtering and Detection: The DTMF signal was filtered through each of the eight bandpass filters using manual convolution implementation. The filtered outputs clearly demonstrated the selectivity of the filter bank—only the two filters corresponding to the pressed key's row and column frequencies produced significant output amplitudes, while the remaining six filters showed minimal response. Energy measurements (peak values) across all filter outputs enabled robust key detection by identifying the two filters with maximum energy.

3. Key Decoding Accuracy: The decoder successfully identified the pressed keys (here is 5) by mapping the detected row and column frequency indices to the DTMF keypad matrix. The time-domain plots of filtered outputs provided visual confirmation that only the relevant frequency components passed through their respective filters, validating the effectiveness of the filter bank design.

DISCUSSION:

SHRAVI BANSAL (24EC10130):

This experiment successfully demonstrated the design, implementation, and analysis of a DTMF (Dual-Tone Multi-Frequency) decoder using manually designed FIR bandpass filters in MATLAB. The core objective was to understand the functionality of DTMF signaling and develop a robust filter bank capable of isolating and identifying the dual-tone frequencies corresponding to telephone keypad presses.

DTMF Signal Synthesis: DTMF signals were successfully generated by summing two sinusoidal components—one from the row frequency group (697, 770, 852, 941 Hz) and one from the column frequency group (1209, 1336, 1477, 1633 Hz). The frequency pairs were carefully chosen to avoid harmonic relationships, ensuring that no frequency is an integer multiple of another and that no sum or difference of frequencies equals any DTMF tone. This fundamental property enables reliable detection without interference.

Manual FIR Bandpass Filter Design: Eight FIR bandpass filters were designed using the windowed impulse

response method. Each filter was constructed by:

- Creating an ideal lowpass filter using a sinc function
- Modulating it to the desired centre frequency using cosine multiplication
- Applying a Hann window to reduce Gibbs phenomenon and improve filter characteristics
- Normalizing the frequency response to unity gain at the passband

The filter design avoided all built-in MATLAB filter functions, demonstrating a deep understanding of the underlying signal processing principles. The filter length ($L = 101$) and bandwidth (50 Hz) were chosen to provide narrow passbands capable of isolating individual DTMF frequencies while maintaining reasonable computational efficiency.

- **Filter Selectivity:** The narrow bandwidth of the designed filters (50 Hz) ensured excellent frequency selectivity, with minimal overlap between adjacent passbands. This is crucial for distinguishing between closely spaced DTMF frequencies.
- **Windowing Impact:** The Hann window effectively reduced sidelobe levels in the frequency response, minimizing potential interference from out-of-band frequencies.

Practical Implications

This experiment demonstrated how digital signal processing techniques enable reliable telecommunications applications. DTMF decoding using FIR filter banks is widely used in telephony systems, automated response systems, and remote-control applications. The skills developed—manual filter design, frequency domain analysis, and signal detection—are directly applicable to broader DSP challenges including audio processing, communication systems, and biomedical signal analysis.

N DEEKSHITHA (24EC10105):

This experiment demonstrates the design and analysis of a Dual-Tone Multi-Frequency (DTMF) encoder and decoder using manually implemented FIR bandpass filters in MATLAB. The goal is to understand how a combination of digital filters can be used to detect telecommunication keypad tones based on their frequency content.

DTMF Signal Synthesis:

Each key press on a telephone keypad generates two sinusoidal tones—one from the low-frequency (row) group and one from the high-frequency (column) group. These tones are summed to form a DTMF signal. The frequencies are carefully chosen to be non-harmonic, ensuring that no tone is an integer multiple of another, which prevents misinterpretation or overlap between signals.

FIR Bandpass Filter Design:

Eight FIR filters were designed using the windowing method (Hann window). The sinc function defines an ideal lowpass filter, which is then modulated to the desired center frequency to obtain a bandpass filter. The use of a Hann window reduces the Gibbs phenomenon and improves the roll-off characteristics.

Parameters such as filter length ($L = 101$) and bandwidth ($BW = 50$ Hz) were selected to provide good frequency resolution.

Frequency Response Observation:

The frequency response plots exhibited sharp peaks at each DTMF tone frequency, confirming that the filters were selective and well-tuned. The extra peaks seen at higher frequencies happened because FIR filters are symmetrical, and they didn't interfere with detection.

Signal Detection and Decoding:

When the generated signal (for key '5') was passed through all filters, only two filters—corresponding to 770 Hz and 1336 Hz—showed high energy outputs. This validated that the decoder can correctly identify the key based on the strongest filter responses.

Applications and Significance:

This technique forms the basis for DTMF decoding used in telephone exchanges, IVR (Interactive Voice Response) systems, and remote control interfaces. The experiment enhanced understanding of filter bank design, frequency response characteristics, and signal detection principles that are also applicable in speech, audio, and communication systems.

Through this, we can get how filter bandwidth and window selection directly affect tone separation accuracy.